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Editors
European Journal of Operational Research

Dear Editors,

Please find enclosed the manuscript *"A beam search for the air cargo palletization problem with weight distribution constraints"*, which I submit for consideration for publication in the European Journal of Operational Research as a Theory and Methodology paper.

The manuscript addresses the palletization of air cargo into unit load devices (ULDs), a three-dimensional multiple bin size bin packing problem with non-cuboid bins, weight capacities, fragile boxes, support constraints and center of gravity constraints. The benchmark and the current state of the art for this problem were both published in EJOR (Paquay et al., 2018; Heßler et al., 2025), and the paper builds directly on this line of work.

The main contributions are the following:

- The adaptation of a block-based beam search (BSG), one of the best approaches for the container loading problem, to ULDs with bevelled corners, a floor edge and a substructure, handling the support and fragility constraints when the blocks are built and placed.
- A treatment of the center of gravity inside the search, through a penalty in the value of the solutions, combined with a final translation of the load.
- A multi-ULD framework whose reduction phase replaces each ULD by at most two smaller ones, which accounts for most of the savings in volume.
- An evaluation on the 300 benchmark instances in the three settings of Heßler et al., with parameters tuned once on a separate training set and a comparison at the same hardware-normalized running time. The algorithm improves on the state of the art in both volume and weight distribution: at a comparable running time it uses 2.9–3.8% less ULD volume, and at most 1% of its ULDs violate the center of gravity constraint, against 7–11% for the previous method; with a few seconds per instance the savings in volume reach 7.1–8.2%.

All the solutions are checked by an independent validator, and the code, the instances, the solutions and the scripts that reproduce the tables and figures are publicly available at https://github.com/rilianx/bsg-uld.

This manuscript is original, has not been published, and is not under consideration for publication

elsewhere. The author declares no competing interests.

Sincerely,

Ignacio Araya
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